

FC21 - <offline>

"STD_DRIVE"Drive Control Module

Name: STD_DFamily: SETEK

Author: GLVersion: 2.1

Block version: 1

Time stamp Code: 09/02/2014 01:11:49 PM

Interface: 01/10/2006 08:58:05 AM

Lengths (block/logic/data): 01288 00900 00050

Name	Data Type	Address	Comment
IN		0.0	
IN	Int	0.0	Item Number (64 bytes offset pr. number)
AUTO_START	Bool	2.0	1=Start, Auto
STOP	Bool	2.1	1=Stop, Auto and Man.
RUNNING_FEEDBACK	Bool	2.2	1=Active (RF OK)
MCC_TRIP	Bool	2.3	1=OK
LOCK	Bool	2.4	1=Locked (Forced to auto)
OUTPUT_TO_STARTER	Bool	2.5	Output to motor starter.
OUT		0.0	
START_REQUEST	Bool	4.0	Request motor start
ASD_ALARM_BIT	Bool	4.1	Alarm is active
IN_OUT		0.0	
TEMP		0.0	
Interf	UDT_D	0.0	
GlobalVars	Struct	40.0	Global variables [DB9]
Allways_0	Bool	40.0	
Allways_1	Bool	40.1	
bit12	Bool	40.2	
Alarm_Ack	Bool	40.3	
Alarm_Reset	Bool	40.4	
Set_all_to_auto	Bool	40.5	
FF_ALARM_RESET	Bool	40.6	
bit17	Bool	40.7	
Pulse_100ms	Bool	41.0	
Pulse_1sec	Bool	41.1	
Pulse_5sec	Bool	41.2	
Pulse_1min	Bool	41.3	
ResetComp_StartStop_time	Bool	41.4	*TEST FUNCTION* Resets all running Start-Start and Stop-Start timers
ProcessVal_Sim	Bool	41.5	*TEST FUNCTION* 1=Enabled, Process val. write-back, Trip + Warn works
GlobalRestart	Bool	41.6	
Feedback_Sim	Bool	41.7	*TEST FUNCTION* 1=Enabled, Motors and Valves ignores FB. Indic. works
GlobalAlarmVars	Struct	42.0	Global alarm variables [DB9] +2bytes
Alarm_acknowledge	Bool	42.0	
Lamp_Bit_1	Bool	42.1	Trigger bit for alarm lamp
New_alarm	Bool	42.2	Trigger bit for alarm horn
FF_Alarm	Bool	42.3	First Fallout Alarm detected
bit120	Bool	42.4	
bit121	Bool	42.5	
bit122	Bool	42.6	
bit123	Bool	42.7	
EMG_ShutDown	Bool	43.0	On EMG disable request signal in valve and drive blocks

Name	Data Type	Address	Comment
bit125	Bool	43.1	
bit126	Bool	43.2	
bit127	Bool	43.3	
bit128	Bool	43.4	
bit129	Bool	43.5	
bit130	Bool	43.6	
bit131	Bool	43.7	
RH_sec	Real	44.0	
RETURN		0.0	
RET_VAL		0.0	

Block: FC21DRIVE CONTROL MODULE

```
*****
*          DRIVE CONTROL MODULE          *
*****
*          ©2009 SETEK Process & Data      *
*          *                               *
*          Part of SETEK S7 Standard library *
*****
* Ver.: 2.1      -      Date: 11.06.2009   *
* Code by: Gregers Lund                    *
*****
Block requires:
DB9  - Word 0 - PLC system - Word
DB9  - Word 2 - PLC alarm - Word
DB21 - SCADA/HMI Drive Control Datablock (DB_DRIVE)
M1.3 - 1_Sec_Pulse

-- History --
v01: Coded all functions according to AGA/Linde specifications
for iFIX interface. Cleared for testing. /GL
v02: Moved Local vars. to main UDT, block only uses one datablock.
v03: Changed Auto/Manual due to wrong description in Linde Standards
v04: Added status for ASD alarm bit output /GL
v05: Changed fault signal to _UZ. Added Feedback_Sim /AL/GL
v06: Added Restart after Fault function, using global Restart signal /GL
v07: Added EMG reset of Request bit /GL
v08: Changed 1 sec. puls to internal bit for valid counting
v2.0: Major revision. New UDT...
V2.0a: Network "Set All to Auto" removed, Network 7 changed, Network 8 added /JJ
V2.1: Changes to Auto/Man switch. Added _Stop indication. Misc. changes. /GL
```

Network: 1

Loads Input Words and transfers states to Local Words/bits

```
// Pointer DB21
L      #IN // Load Item Number          #IN          -- Item Number (64 bytes offset
                                         pr. number)

L      40 // Offset i DB
*I
SLW    3  // Shift Left 3 bits to format pointer correctly
LAR1
```

Network: 2

Transfers states from command words to internal words/bits

```
OPN    "DB_MISC_FUNCTIONS"                DB9
L      DBW    0
T      LW     40          // PLC system word

OPN    "DB_MISC_FUNCTIONS"                DB9
L      DBW     2
T      LW     42          // PLC alarm word

// DB21
OPN    "DB_D"                            DB21
// ----- BOOL -----
L      DBD [AR1,P#0.0]
```

```
T      LD      0

// ----- REAL -----
L      DBD [AR1,P#4.0]
T      #Interf._Kih                                #Interf._Kih      -- Running hours Hours

L      DBD [AR1,P#8.0]
T      #Interf._Kim                                #Interf._Kim      -- Running hours Minutes

// ----- INTEGER -----
L      DBW [AR1,P#24.0]
T      #Interf._DelaySP                            #Interf._DelaySP  -- Running feedback alarm time SP

L      DBW [AR1,P#26.0]
T      #Interf._DelayCnt                            #Interf._DelayCnt -- Running feedback alarm time counter

// ----- WORD -----
L      DBW [AR1,P#38.0]
T      LW      38
```

Network: 3Status Auto/Manual

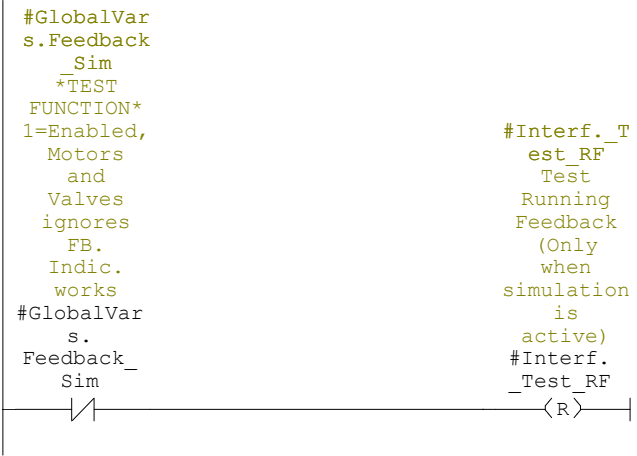
```
A      #Interf._Test_RF // If RF test is active, the drive Lock is overruled #Interf._Test_RF -- Test Running Feedback (Only when simulation is active)
JC      AM01

A      #LOCK                                #LOCK      -- 1=Locked (Forced to auto)
=      #Interf._L                                #Interf._L      -- Motor Locked
JCN     AM01 // If drive is locked, no change

SET
R      #Interf._MA                                #Interf._MA      -- Motor Man/Auto select

AM01: NOP 0
```

Network: 4Reset RF test function when simulation is off



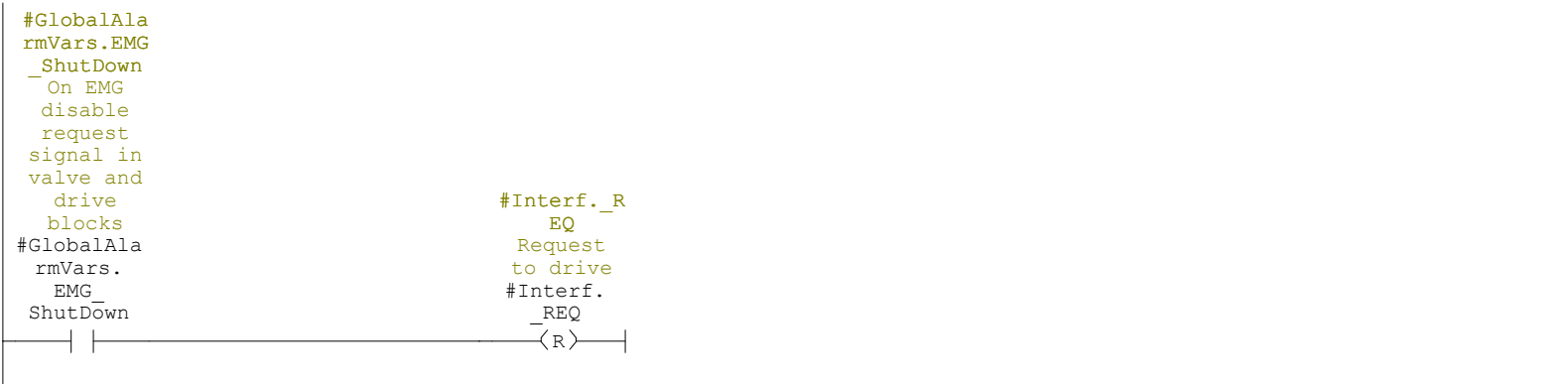
Network: 5Set all to auto function



Network: 6	Internal 1sec. pulse to ensure correct counting
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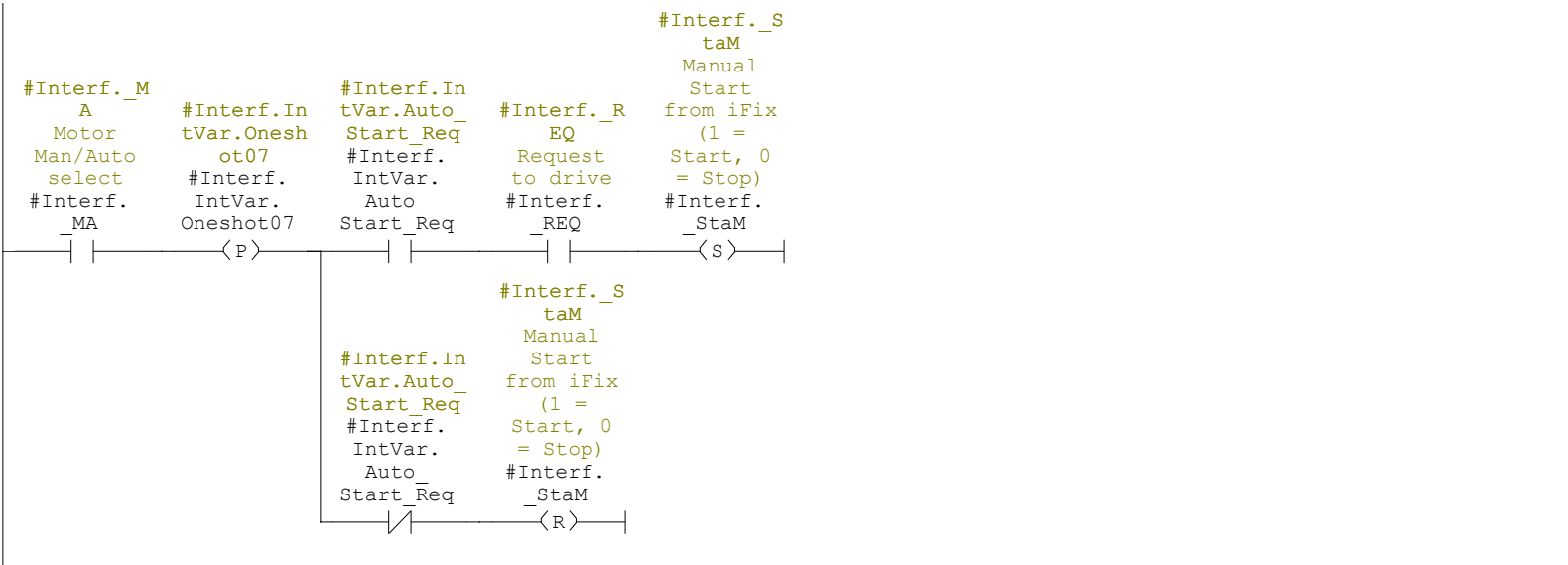


Network: 7	EMG Reset of REQ. bit
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```
Network: 8      Set manual state to the same as auto state
```

```
100830GL/ Removed #START_REQUEST to ensure correct change from Auto to Manual
```

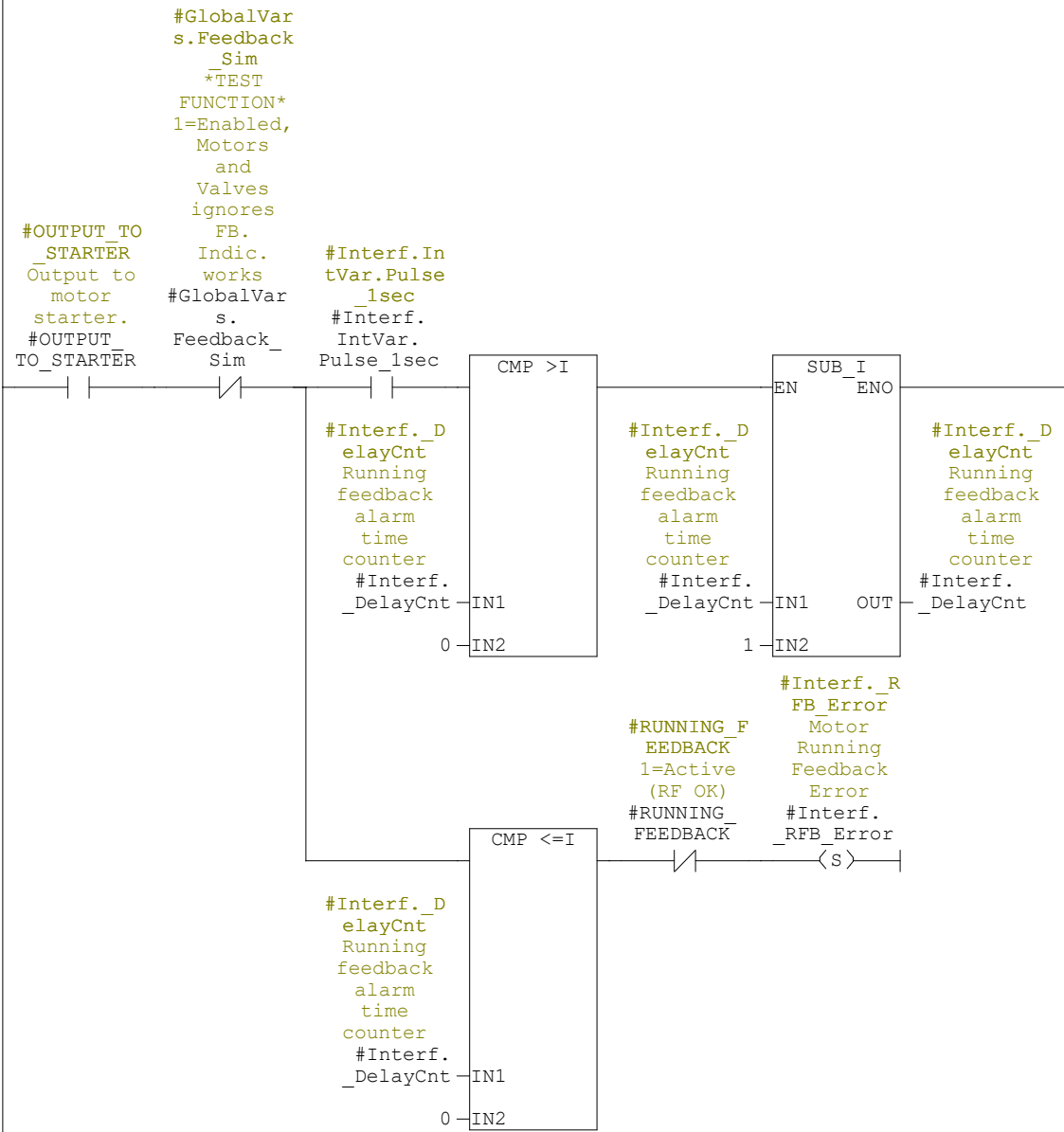


Network: 9#Interf.IntVar.Auto_Start_Req is checked before #Interf._MA

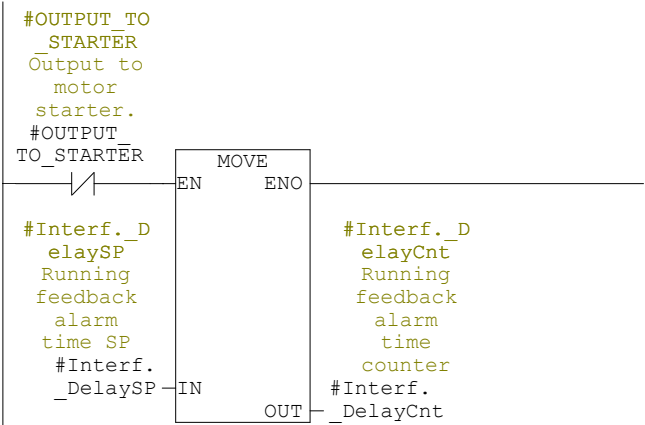
To ensure that #Interf._StaM is set to the correct state if #Interf._MA and #Interf.IntVar.Auto_Start_Req is activated in the same step in a sequence.



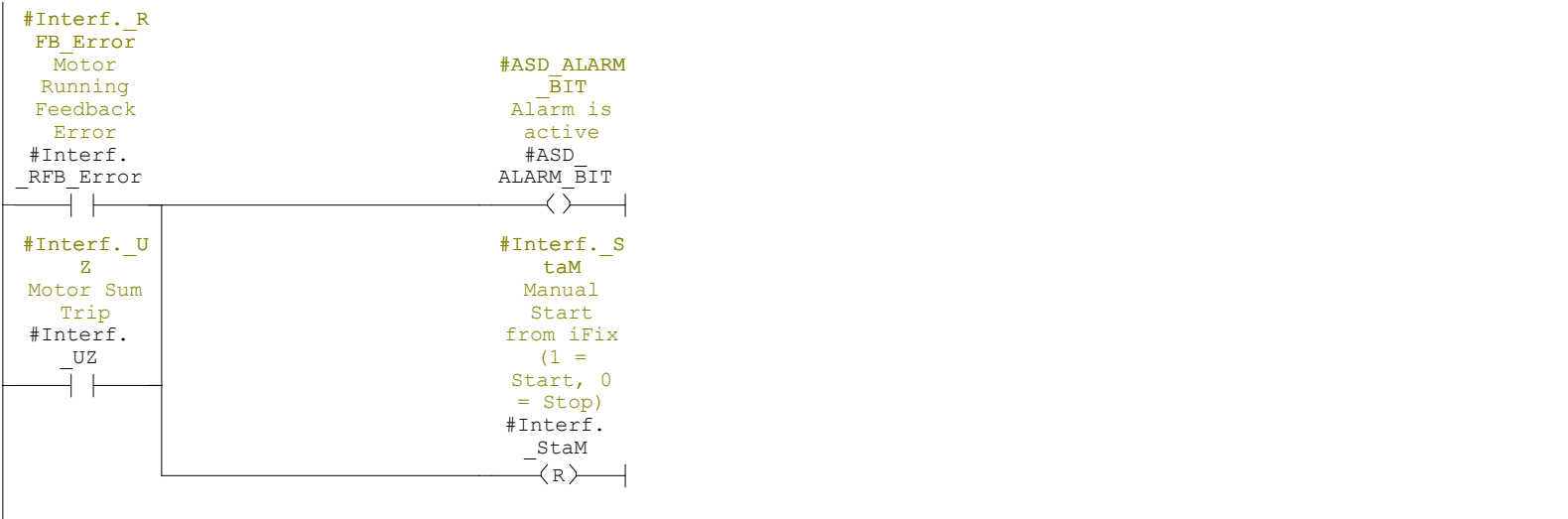
Network: 10Running feedback alarm



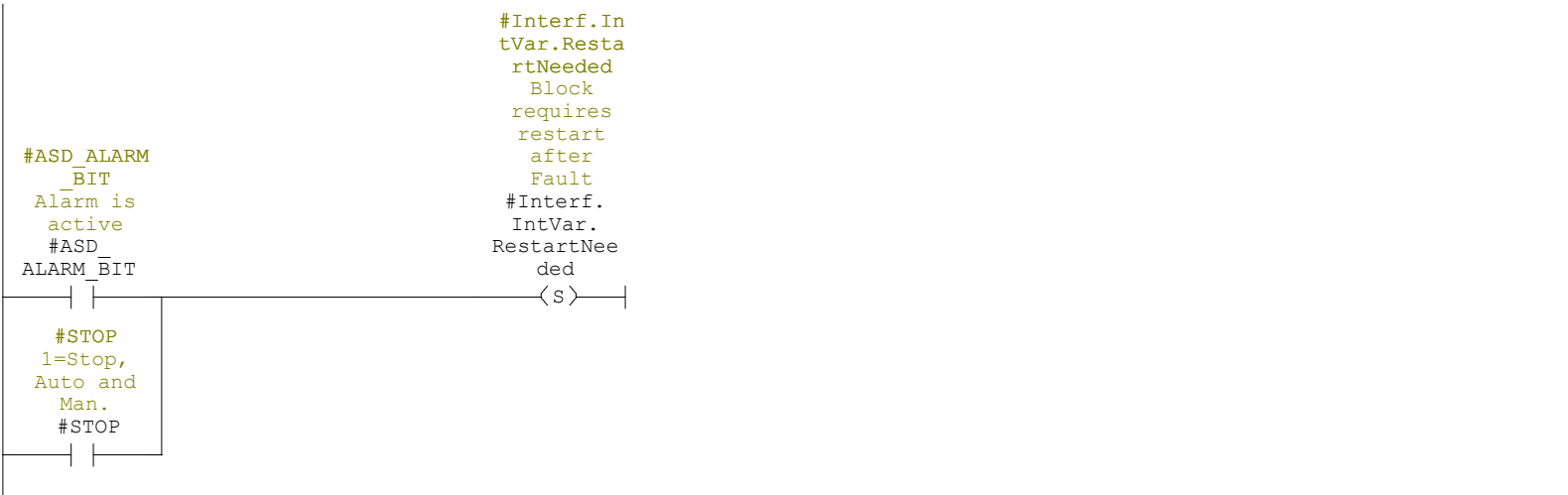
Network: 11Reset feedback delay



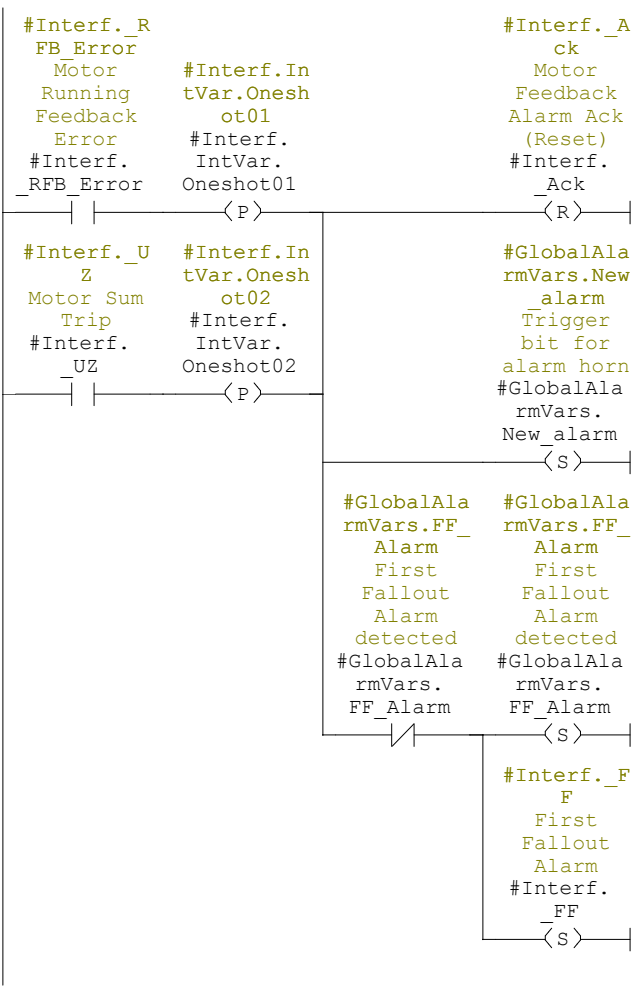
Network: 14	ASD alarm bit status
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```
Network: 15      Restart Needed is set when block in fault or in stop
```



Network: 16New alarm and set FF global and local if not detected before



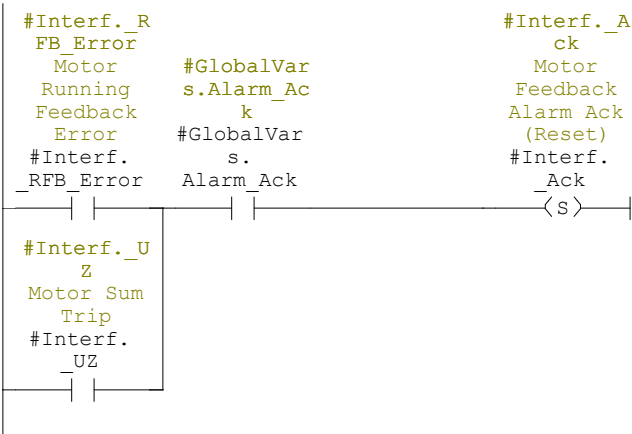
Network: 17Reset FF Alarm



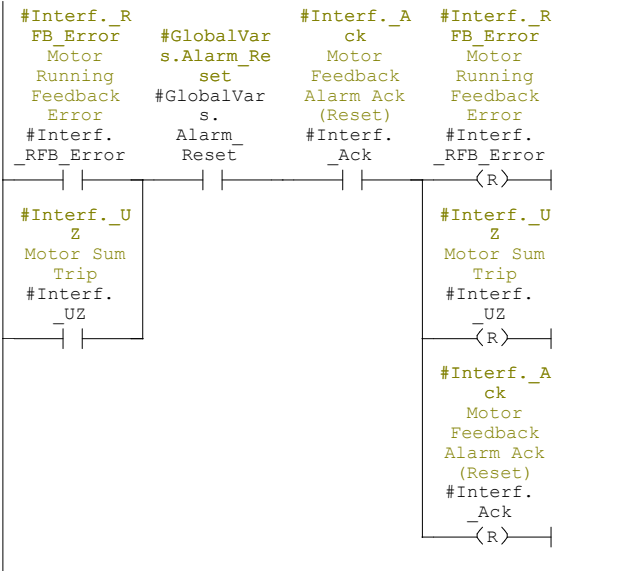
Network: 18Active alarm



Network: 19Alarm ACK

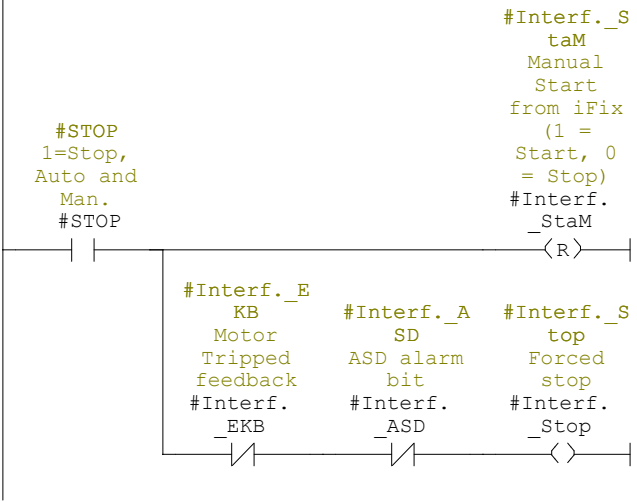


Network: 20Reset alarms

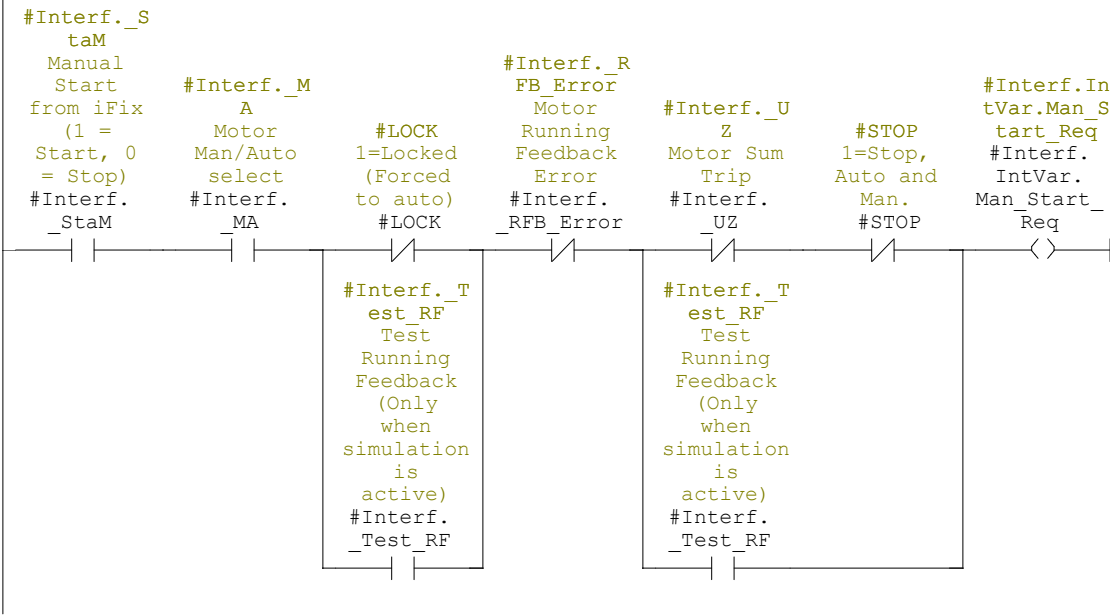


Network: 21Reset Manual Start when Stop is activated

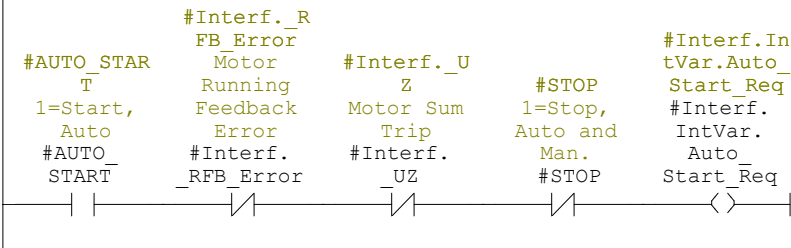
100830GL/ Added _Stop indication



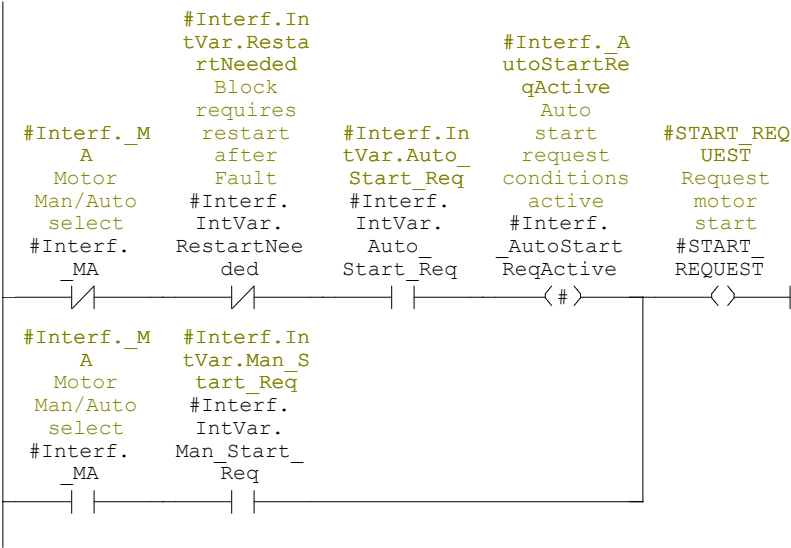
Network: 22Manual start request



Network: 23Auto start request



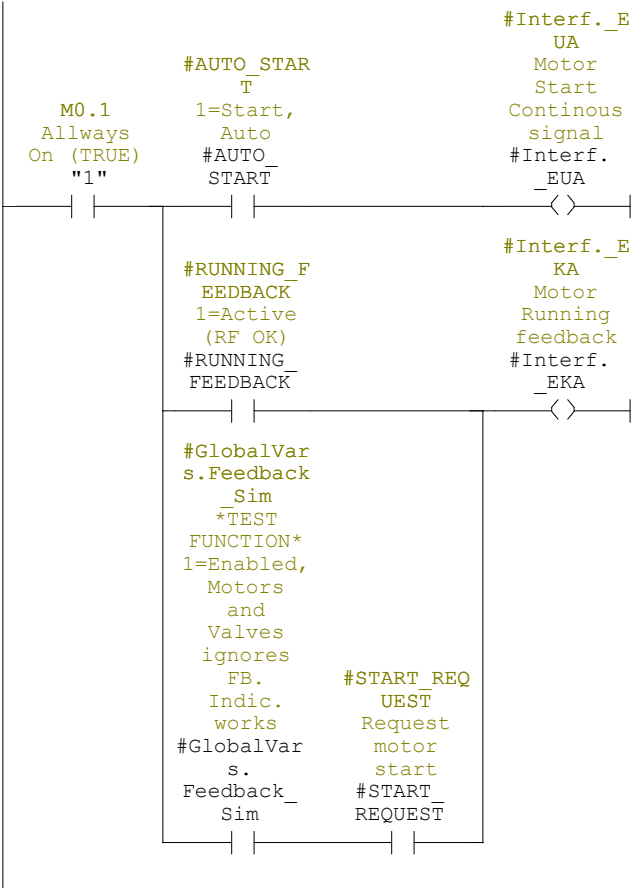
Network: 24Start request



Network: 25Reset control commands



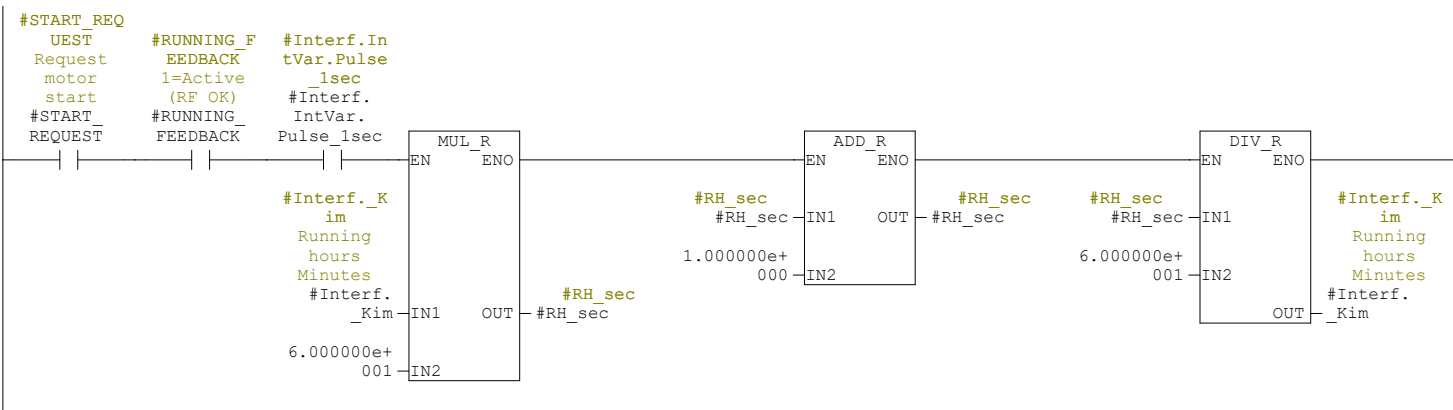
Network: 26Status signals



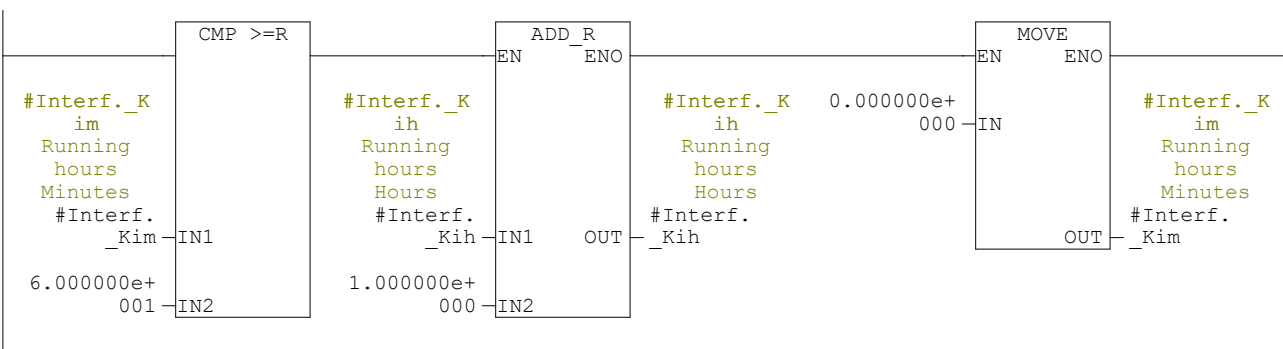
Network: 27Reset FF Alarm



Network: 28
Calc. running minutes



Network: 29
Calc. runnig hours



Network: 30
Transfer states from internal words/bits to command words

```

OPN  "DB_MISC_FUNCTIONS"                                DB9
L    LW  42
T    DBW  2      // PLC alarm word

// DB21
L    #IN          // Load Item Number                    #IN
L    40           // Offset i DB                          offset pr. number)
*I
SLW  3           // Shift Left 3 bits to format pointer correctly
LAR1

OPN  "DB_D"                                              DB21

// ----- BOOL -----
L    LD  0
T    DBD [AR1,P#0.0]

// ----- REAL -----
L    #Interf. Kih          #Interf._Kih      -- Running hours Hours
T    DBD [AR1,P#4.0]

L    #Interf. Kim          #Interf._Kim      -- Running hours Minutes
T    DBD [AR1,P#8.0]

// ----- INTEGER -----
L    #Interf._DelayCnt      #Interf._DelayCnt -- Running feedback alar
T    DBW [AR1,P#26.0]      m time counter

// ----- WORD -----
```

L

LW38

T

DBW [AR1,P#38.0]